

On the Folding of Sheets

How quires, signatures and folios shaped the book

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ABSTRACT

A sheet folded once gives a folio; twice, a quarto; three times, an octavo. We review how these gatherings, called quires, determined the size and structure of books, and how their arithmetic survives in modern imposition.

KEYWORDS bookbinding · imposition · typography

§ 1

Introduction

Before a book is sewn, its printed sheets are folded into *quires* (or gatherings). A sheet folded n times gives 2^n leaves and 2^{n+1} pages, a fact that every printer had to master, since the pages of a quire are not printed in reading order. Section 2 derives the order; Table 1.1 summarizes the classical formats.

FORMAT	FOLDS	LEAVES	PAGES
Folio	1	2	4
Quarto	2	4	8
Octavo	3	8	16

TABLE 1.1. Classical book formats.

§ 2

Imposition

2.1 The arithmetic of folding

DEFINITION 2.1. A *quire* of n folds is a sheet folded n times, giving $L = 2^n$ leaves and $P = 2L$ pages.

Pages facing each other on the same side of the unfolded sheet add up to a constant:

$$p + p' = P + 1. \quad (2.1)$$

PROPOSITION 2.2. *In a quire of P pages, page p and page $P + 1 - p$ are printed side by side.*

Proof (Proposition 2.2). Folding preserves adjacency of the outer edges; unfolding the innermost fold first, induction on n gives Equation (2.1). $\square$

2.2 Computing the layout

The order of pages on each side of the sheet can be computed directly:

```
IMPOSE.PY
1 def pairs(pages: int) → list[tuple[int, int]]:
2     """Pages printed side by side in a quire."""
3     return [(p, pages + 1 - p) for p in range(1, pages // 2 + 1)]
```

Remark 2.3. Modern imposition software, e.g. [Typst](#)-based workflows, still relies on this rule.

§ 3

Conclusion

The quire is the atom of the codex; its arithmetic, sketched in Section 2 (IMPOSITION), explains formats that are still named after it [1].

Bibliography

[1] D. E. Knuth, *The TeXbook*. Addison-Wesley, 1984.